CVaR Portfolio Optimization Report

Minimizing Tail Risk in Low-Risk Portfolios

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1 Summary

This report looked at different ways to manage portfolio risk using stock data from 2019 and 2020, focusing on Conditional Value-at-Risk (CVaR) to measure potential losses. We first optimized portfolios to minimize daily average CVaR at different confidence levels. We found that higher confidence levels tended to concentrate the portfolio in a few "core" stocks, reducing diversification and making the portfolio less likely to perform well in future periods. When we tested the 2019-optimized portfolios on 2020 data, the out-of-sample CVaR was higher than expected, especially during periods of market stress (i.e. COVID), showing that past patterns don't always predict future results.

We also tried a conservative approach that minimizes the worst monthly CVaR in historical data. This slightly smoothed risk in some months but didn't meaningfully reduce extreme losses in 2020, suggesting that historical worst cases may not reliably indicate future ones. Finally, we applied a rolling monthly re-optimization, updating the portfolio each month using the prior year's data. This reduced average monthly CVaR by about 30% compared to keeping the 2019 portfolio static, though it caused large changes in portfolio weights from month to month.

Overall, we recommend regularly updating portfolio allocations, starting with monthly updates and possibly moving to shorter windows, to stay aligned with current market trends while avoiding reliance on outdated data. We suggest to be cautious when choosing very high CVaR confidence levels, as this can lead to highly concentrated portfolios that may not perform well in future market conditions if the few core stocks happen to underperform.

2 Introduction

2.1 Portfolio Optimization

In portfolio optimization, we are often more interested in risk minimization subject to a desired rate of return, rather than the more rudimentary approach of attempting to maximize profit, especially when managing money for investors who have their own expectations of acceptable risk. Portfolio managers also have fiduciary duties and legal obligations, including requirements under SEC regulations, to ensure that portfolios are managed within reasonable levels of risk. Maximizing profit alone is not an especially meaningful problem, since it ignores the variability of returns and the possibility of large losses. For example, one could select the highest performing stock from the previous year and allocate the entire portfolio to it, a strategy that would almost certainly fail in practice. The problem of risk minimization is much more intricate: given a desired return, how can we minimize the risk we take on to get there? This requires us to define a measure of risk.

2.2 Measures of Risk

One such definition we could use is Value-at-Risk (VaR), which, in essence, describes the maximum loss that we expect not to exceed at a given confidence level over a trading period. At confidence level β, the VaR corresponds to the $(1-\beta)th$ percentile of the portfolio loss distribution for a particular investment allocation. This measure has some limitations that we do not discuss here for the sake of brevity. Instead, we focus on a similar measure of risk: Conditional Value-at-Risk (CVaR). This describes the average losses beyond the VaR, making it capable of describing the severity of the loss, something particularly important when the distribution of returns is highly irregular.

2.3 Project Objectives

In this report, we use stock price data from 2019 to construct risk-optimal portfolio allocations for 2020 at a target daily return of 0.02%. The 2019 data is used to determine the portfolio weights, while the 2020 data allows us to evaluate out-of-sample performance. We explore several strategies, including minimizing daily average CVaR at a 95% confidence level, examining the effect of different confidence levels, and implementing a conservative approach that minimizes the maximum monthly CVaR to control extreme losses. Additionally, we investigate the benefits of monthly portfolio re-optimization and assess the stability of the resulting allocations over time. The goal of these analyses is to evaluate how different risk management strategies perform under varying market conditions and to provide actionable recommendations for constructing low-risk portfolios.

3 Methodology

3.1 Data Preprocessing

We use daily stock price data for 100 stocks from the Nasdaq index in 2019. The data is provided in CSV format, with a date column that we convert to a `datetime` object and set as the index, followed by 100 columns each corresponding to a single stock, and an additional column for the NDX index which we drop from the dataframe. From the price data, we calculate daily percentage returns for each stock. These returns are used both to estimate expected returns and to construct the loss distribution required for CVaR optimization.

3.2 CVaR Model

To construct a risk-optimal portfolio using CVaR, we build a linear program that minimizes the portfolio CVaR subject to a required target return R, using an auxiliary variable u_k to

linearize the piecewise max function. We also use the constraints that weights must sum to 1 (weights are a proportion of our investment budget) and that weights are non-negative.

Then we seek to minimization the objective:

$$\tilde{F}_\beta(\mathbf{x}) = \min_{\alpha,u} \alpha + \frac{1}{(1-\beta)q} \sum_{k=1}^{q} u_k$$

Where $u_k \geq -\mathbf{x}^T \overline{\mathbf{y}}_k - \alpha$ and $u_k \geq 0$.

With the additional minimum return constraint that $\mathbf{x}^T \overline{\mathbf{y}}_k \geq R$, so that we are only considering portfolios that meet the required rate of return. We might note that the optimally safe portfolio may have a return greater than R, which would be a welcome outcome.

Here, each u_k represents the loss in excess of the VaR threshold α. We sum these excess losses, divide by the proportion of observations in the $(1-\beta)$ tail to compute the average, and then add back the VaR α, which was initially subtracted when calculating the excess losses. Optimizing this objective finds the weights (proportion of investing capital) to invest in each stock that correspond to the minimum CVaR for the portfolio at confidence level β, while still meeting the required rate of return.

3.3 Optimization

Just to illustrate, we define a function to optimize a portfolio by:

```
def optimize_cvar_portfolio(daily_returns, average_daily_returns,
    beta = 0.95, min_return = 0.0002):
...
    # Set objective: minimize CVaR
    cvarMod.setObjective(alpha + (1/((1 - beta)*n_scenarios)) * gp.
        quicksum(u[k] for k in range(n_scenarios)), sense = gp.GRB.
        MINIMIZE)
...
```

And the key auxiliary constraint:

```
...
    # Auxiliary constraint
    for k in range(n_scenarios):
        portfolio_return_k = gp.quicksum(cvarModX[i] *
            returns_matrix[k, i] for i in range(n_stocks))
        cvarMod.addConstr(u[k] >= -portfolio_return_k - alpha)
...
```

Where we've omitted the other constraints, again for brevity.

3.4 Model Evaluation on 2020 Stock Data

We can identify the returns for the 2020 by taking the dot product of the optimal weights and the daily returns:

$$\mathbf{x}^{*T}\mathbf{y}_{2020}$$

Which we then find the CVaR by sorting, taking the bottom $1 - \beta$ percentile, and finding the average:

```
def calculate_cvar(returns, beta = 0.95):
    sorted_returns = np.sort(returns)
    var_index = int((1 - beta) * len(sorted_returns))
    cvar = -sorted_returns[:var_index].mean()
    return cvar
```

3.5 Minimizing the Maximum Monthly β-CVaR

We achieve this by grouping the data by month.

```
def optimize_conservative_cvar_portfolio(daily_returns,
   average_daily_returns, beta = 0.95, min_return = 0.0002):

    # Prepare monthly grouping
    daily_returns_copy = daily_returns.copy()
    daily_returns_copy['Year'] = daily_returns_copy.index.year
    daily_returns_copy['Month'] = daily_returns_copy.index.month
    monthly_groups = daily_returns_copy.groupby(['Year', 'Month'])
...
```

The optimizer then minimizes the largest monthly CVaR, effectively controlling the worst case loss. This makes the model slightly better in the worst cases, at the cost of potentially reducing performance for the average cases.

```
...
    # Variables for each month's CVaR calculation
    alpha_months = cvarMod_conservative.addMVar(n_months, lb = -gp.
       GRB.INFINITY)
    u_months = []
    for m in range(n_months):
        n_days_month = len(monthly_returns_list[m])
        u_months.append(cvarMod_conservative.addMVar(n_days_month,
           lb = 0))
...
```

3.6 Monthly Portfolio Re-optimization

To prepare the data for monthly re-optimization, we clean it as in Section 3.1, this time concatenating the 2019 and 2020 stock data. For each month in 2020, we take the previous 12 months of data, solve the CVaR minimization, and apply the resulting portfolio to that month. In this setup, the training set is updated each month by adding the most recent month and removing the data from 13 months prior.

Rolling lookback window:

```
...
    train_end = pd.to_datetime(f'{year}-{month:02d}-01') - pd.
       offsets.Day(1)
    train_start = train_end - pd.DateOffset(years=1) + pd.offsets.
       Day(1)
    lookback_data = daily_returns_combined.loc[train_start:train_end
       ].dropna()
...
```

Optimizing over each window:

```
...
    avg_returns = lookback_data.mean()
    weights, _ = optimize_cvar_portfolio(lookback_data, avg_returns)
    monthly_optimal_weights[f"{year}-{month:02d}"] = weights
...
    portfolio_returns = (test_data * weights).sum(axis = 1)
...
```

We then compute summary statistics over the set of 12 CVaRs from 2020 using standard NumPy functions.

3.7 Portfolio Stability

To assess portfolio stability, we compute the absolute month-to-month change in each portfolio weight. If any weight changes by more than 0.05, the portfolio is classified as unstable.

4 Results

4.1 Optimal Portfolio Weights Using Minimal Daily Average CVaR

Using the 2019 stock data, we found the following weights by minimizing CVaR at $\beta = 0.95$:

Optimal Portfolio Weights				**Metrics**	
Stock	Weight	Stock	Weight	Metric	Value
XEL	30.39%	CSX	3.43%	2020 Average Daily CVaR	4.75%
CHTR	26.50%	COST	3.04%	Expected Daily Return	0.1227%
AMGN	8.08%	EBAY	2.35%		
CTXS	6.56%	JD	1.98%		
CHKP	6.27%	CTSH	1.96%		
EXC	5.22%	LULU	0.18%		
KHC	4.05%				

Table 1: Optimal portfolio allocation at $\beta = 0.95$ (left) and portfolio risk and return metrics for 2020 (right).

4.2 Portfolio Optimization at $\beta = 0.9$ and $\beta = 0.99$

We did the same, $\beta = 0.9$ and $\beta = 0.99$ to determine the effect on portfolio allocation by changing the risk level:

Optimal Portfolio Weights				**Metrics**	
Stock	Weight	Stock	Weight	Metric	Value
CHTR	24.06%	EXC	6.43%	2020 Average Daily CVaR	3.2775%
XEL	21.30%	WBA	5.29%	Expected Daily Return	0.1216%
MDLZ	13.66%	EBAY	4.87%		
AMGN	4.80%	CTXS	4.73%		
ORLY	4.20%	PEP	3.16%		
DLTR	3.07%	CHKP	1.83%		
KHC	1.47%	DXCM	0.48%		
QCOM	0.37%	CPRT	0.29%		

Table 2: Optimal portfolio allocation at $\beta = 0.9$ (left) and portfolio risk and return metrics for 2020 (right). Training (2019 stock data) CVaR = 0.0089

Optimal Portfolio Weights				Metrics	
Stock	Weight	Stock	Weight	Metric	Value
XEL	44.70%	KHC	6.73%	2020 Average Daily CVaR	9.8043%
CHTR	24.81%	NTES	5.10%	Expected Daily Return	0.1307%
CTXS	8.91%	SPLK	4.66%		
AMGN	3.63%	DXCM	0.82%		
SIRI	0.65%				

Table 3: Optimal portfolio allocation at $\beta = 0.99$ (left) and portfolio risk and return metrics for 2020 (right). Training (2019 stock data) CVaR = 0.0125

4.3 Minimizing the Maximum Monthly β-CVaR

The portfolio weights that minimize the worst-case monthly CVaR are:

Conservative Portfolio Weights				Metrics	
Stock	Weight	Stock	Weight	Metric	Value
XEL	45.03%	AMGN	3.54%	2020 Average Daily CVaR	4.8368%
CHTR	24.35%	SPLK	4.69%	Expected Daily Return	0.1303%
CTXS	8.97%	NTES	5.11%		
KHC	6.73%	DXCM	0.84%		
SIRI	0.74%				

Table 4: Conservative risk-optimal portfolio minimizing the maximum monthly CVaR ($\beta = 0.95$). Maximum monthly CVaR: 0.01245.

We can compare the two strategies month-wise to see if we really are minimizing the worst month's risk:

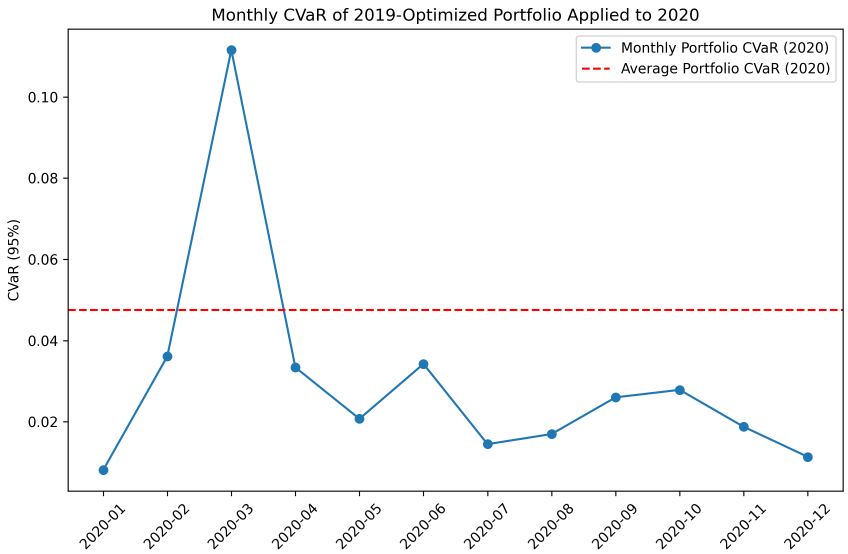

Figure 1: Month-wise CVaR using 2019 weights from average daily CVaR Optimization.

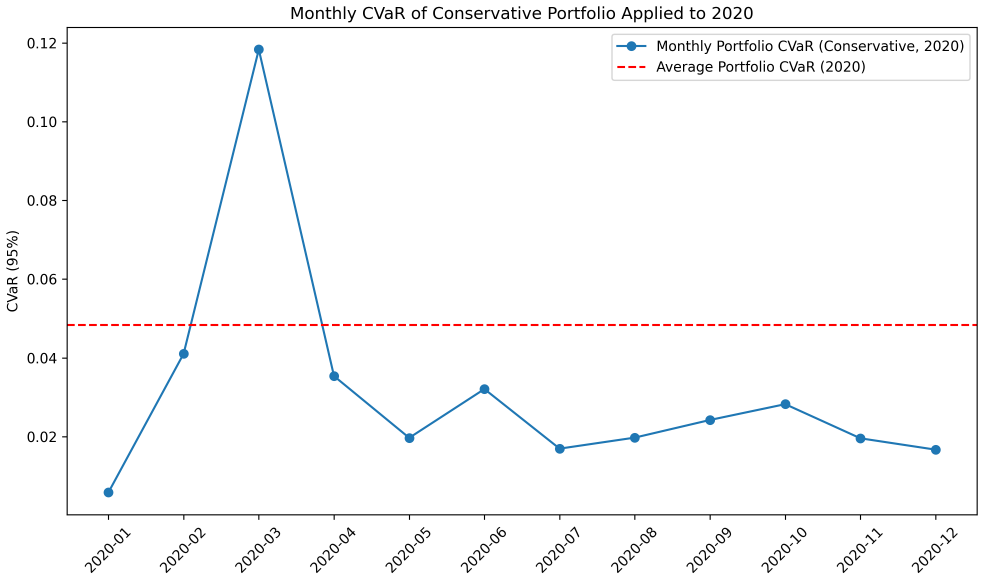

Figure 2: Month-wise CVaR using 2019 weights from minimizing monthly maximum CVaR.

4.4 Monthly Re-optimization

We omit the 12 sets of weights found using monthly re-optimization. Instead, we have summary statistics over the monthly CVaRs:

Rebalanced Portfolio in 2020 Performance (Monthly Re-Optimization)	
Summary Statistics	
Average Monthly CVaR	3.2887%
Std Dev Monthly CVaR	2.3372%
Minimum Monthly CVaR	0.8140%
Maximum Monthly CVaR	10.3875%
Individual Months	
2020-01	0.8140%
2020-02	3.5750%
2020-03	10.3875%
2020-04	4.2191%
2020-05	2.9995%
2020-06	2.8972%
2020-07	3.4080%
2020-08	1.6252%
2020-09	3.4492%
2020-10	2.3339%
2020-11	2.1530%
2020-12	1.6029%

Table 5: Monthly CVaR performance of the rebalanced portfolio in 2020 with monthly re-optimization.

And we visualize the monthly CVaRs:

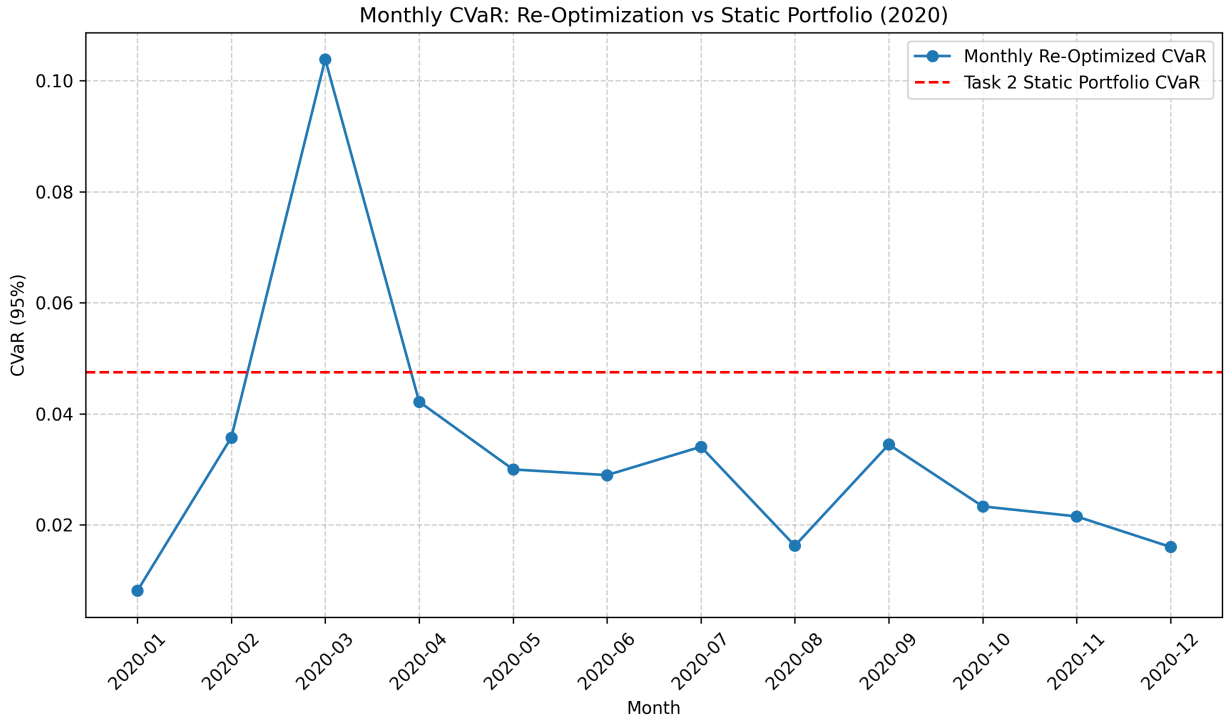

Figure 3: Monthly CVaRs of the conservative portfolio applied to 2020.

4.5 Portfolio Stability

Month	Max Weight Change	Unstable
2020-02	2.35%	No
2020-03	8.58%	Yes
2020-04	45.17%	Yes
2020-05	20.43%	Yes
2020-06	19.29%	Yes
2020-07	8.24%	Yes
2020-08	13.05%	Yes
2020-09	29.00%	Yes
2020-10	19.59%	Yes
2020-11	6.84%	Yes
2020-12	0.00%	No

Table 6: Month-by-month maximum weight changes and portfolio stability.

5 Discussion

5.1 Optimizing Daily Average CVaR

Unsurprisingly, decreasing β led to a lower average daily CVaR. This is because at higher βs, we are averaging over increasingly extreme losses. We may have predicted that operating with higher risk-aversion would reduce our expected returns, but the expected daily return for all three strategies was highly similar. In fact, it was highest for $\beta = 0.99$, although only by 0.008% compared to $\beta = 0.95$. This may be inconsequential, since the optimizer need only meet the required rate of return. More interestingly, as we increase β, the portfolio allocation seems to allocate more heavily in a core of possible "stable" stocks, particularly XEL and CHTR, as opposed to the conventional wisdom that diversifying stocks is safer. It is also important to note that the required rate of return was exceptionally low, lower than the rate of inflation, which is unlikely to be a real target for any wealth management target. The observed differences between risk levels may change under a higher required return.

5.2 Minimizing Maximum Monthly CVaR

Applying this strategy to the 2020 data resulted in a CVaR of 4.84% at $\beta = 0.95$, slightly higher than the original daily-average CVaR strategy, which had 4.75%. This shows that minimizing the maximum monthly risk based on past data does not necessarily reduce exposure to future worst-case losses. Visualizing monthly CVaRs reveals that both strategies follow very similar patterns. The conservative approach slightly smooths CVaR changes in the later months, but neither strategy prevents the sharp spike in February 2020. This suggests that this type of month-based optimization may have limited benefit, particularly when extreme losses are caused by specific events rather than ongoing trends.

5.3 Monthly Re-optimization & Portfolio Stability

5.4 Monthly Re-optimization

The portfolio was re-optimized each month using a rolling window of the previous 12 months of data. This approach was able to improve the average monthly CVaR to 3.29% from 4.75% at β=0.95. Although the raw change is only 1.46%, in relative terms this corresponds to a reduction of about 30.7%. This indicates that using a rolling optimization may be beneficial. Considering that the time to compute a new optimization is mere seconds, we may want to consider even more frequently updating strategies. In practice, trading firms may be optimizing every day, every hour, or even every few seconds, often over shorter and shorter windows.

5.5 Portfolio Stability

From Table 6, we can see that nearly every month was unstable by our definition of instability being a weight change of 5%, with the largest change happening in April, with a single weight change of over 45%. This seems to contrast with the improved CVaR we observed, suggesting that stable portfolio weights are not necessarily correlated with reduced risk.

We may enforce stability by creating a new set of constraints, forcing the optimizer to find weights that move no more than 5% month to month:

$$-0.05 \;\leq\; w_{i,t+1} - w_{i,t} \;\leq\; 0.05$$

Which we could streamline using slack variables:

$$\begin{aligned} w_{i,t+1} - w_{i,t} &\leq 0.05 + s^{+}_{i,t}, \\ w_{i,t} - w_{i,t+1} &\leq 0.05 + s^{-}_{i,t}, \\ s^{+}_{i,t},\; s^{-}_{i,t} &\geq 0. \end{aligned}$$

Where the variables are defined as follows: $w_{i,t}$ represents the weight of stock i in the portfolio at month t, while $w_{i,t+1}$ is the weight in the following month. The slack variables $s^{+}_{i,t}$ and $s^{-}_{i,t}$ allow the constraints to be softened slightly, ensuring the optimizer can still find a feasible solution even if a strict 5% change limit would otherwise be violated. Specifically, $s^{+}_{i,t}$ measures any excess above the upper bound of a weight increase, and $s^{-}_{i,t}$ measures any excess above the upper bound of a weight decrease. By including these slack variables in the optimization, we can enforce approximate stability while maintaining flexibility for the optimizer to adjust the portfolio in response to changing market conditions.

6 Conclusion

We found that at very high βs, the optimization seems to "un-diversify", suggesting that increasing the β to too high may actually be detrimental for generalization to future data. Further, we found that minimizing the worst-case losses in historic data did not improve risk protection in future data, likely because the worst cases are driven by events that simply exist outside of trends in data.

Based on our results, we recommend adopting a frequent re-optimization strategy, starting with monthly updates but also exploring daily or even more frequent re-optimizations over shorter rolling windows. This approach can capture the relevant trends driving stock returns while avoiding reliance on outdated information from months prior.